

Data Analysis

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Electronics Engineering

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Semiconductor Sector Data Analysis

Motivation

- Rise of AI may have changed market structure.
- AI-focused firms differ in growth and risk from legacy semiconductor groups.
- How the semiconductor industry changed after the AI surge.
- Why this matters for investors and policymakers.
- How does this affect us ?

Dataset Overview

- **Dataset Name** : Semiconductor stocks and the AI Surge
- **Source** : Kaggle
- **Time Period** : January 2015 – August 2025 (Filtered from 2012 for relevance).
- **Sampling Frequency**: Daily (Trading Days).
- **Key Variables**: Close (Price), Volume (Liquidity), Stock name.
- **Observations**: 42606 rows (after filtering) & 9 columns.

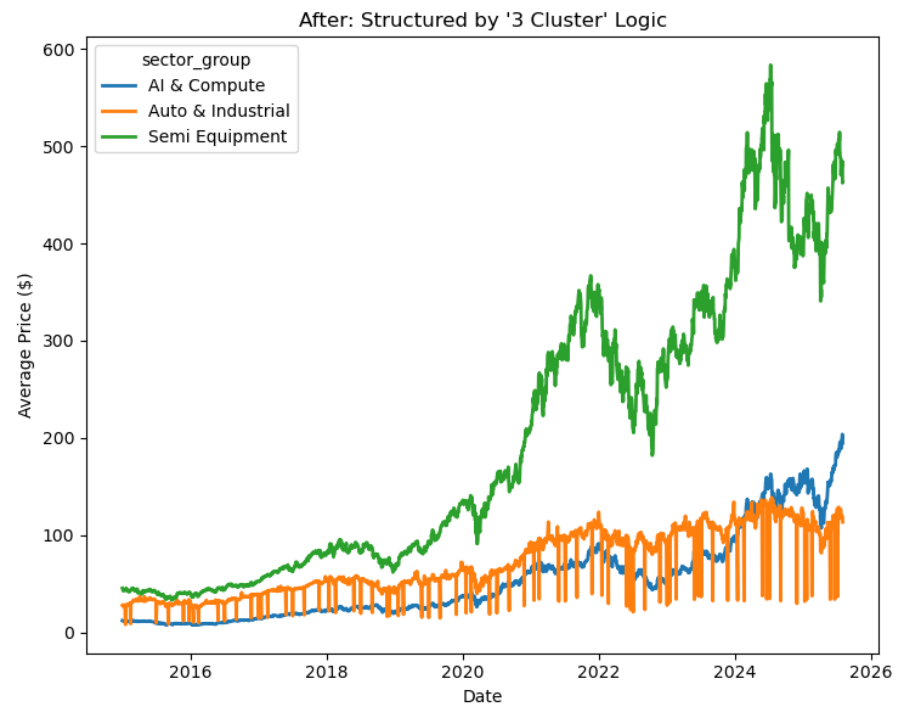
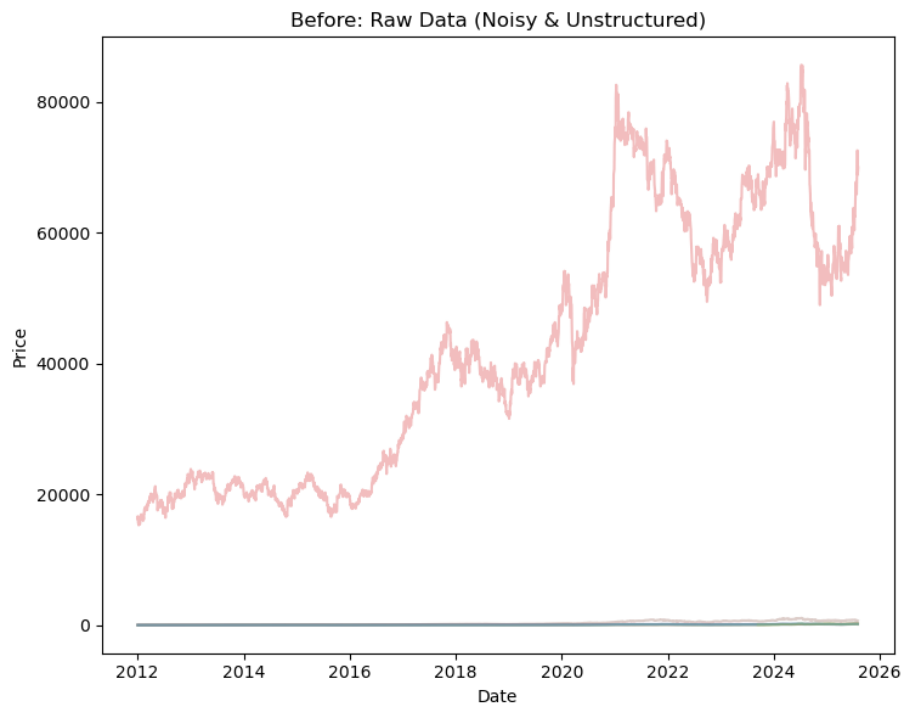
Analyses Performed

- **Visualization and Exploratory Data Analysis (EDA)**
- **Probability and Event Analysis**
- **Statistical Theory Application**
- **Regression and Predictive Modelling**
- **Dimensionality Reduction**
- **Hypothesis Testing**

Data Quality & Preprocessing

- Found minor gaps during non-trading days and dropped them.
- Removed ghost columns.
- Focused on 2015+ to capture the modern GPU era.
- Segmented the industry into three : **AI & Compute** (e.g., NVDA), **Semi Equipment** (e.g., ASML), and **Auto & Industrial** (e.g., NXP).

Before vs After Comparison



The Day that Left a Mark : AI Boom

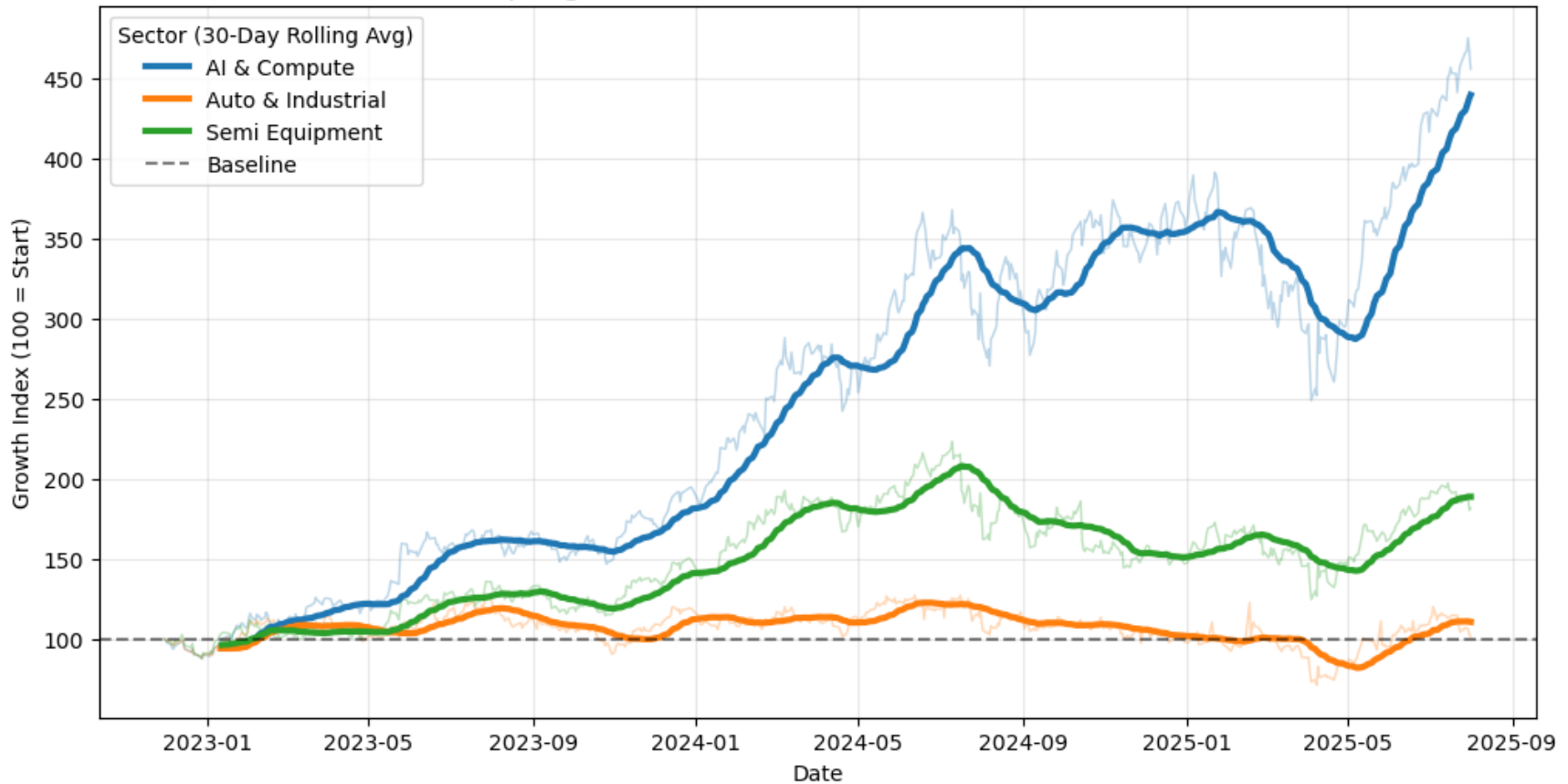
- ChatGPT officially released by OpenAI on **November 30, 2022**.
- Marked a visible inflection point in AI adoption and compute demand.
- Sudden increase in demand for GPUs and advanced accelerators.



Visualization and Exploratory Analysis

- Normalized prices to 100 (fair cross-company comparison) starting **Nov 30, 2022**.
- Evidence of a regime shift, not just a temporary spike.
- Main Observation:
 - > AI & Compute : >250
 - > Auto & Industrial : ~110

The "AI Decoupling": Sector Growth Since ChatGPT (Nov 2022 = 100)

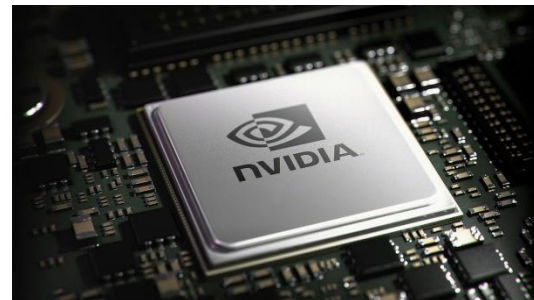


THE DIVERGENCE PLOT



The Rise of NVIDIA®

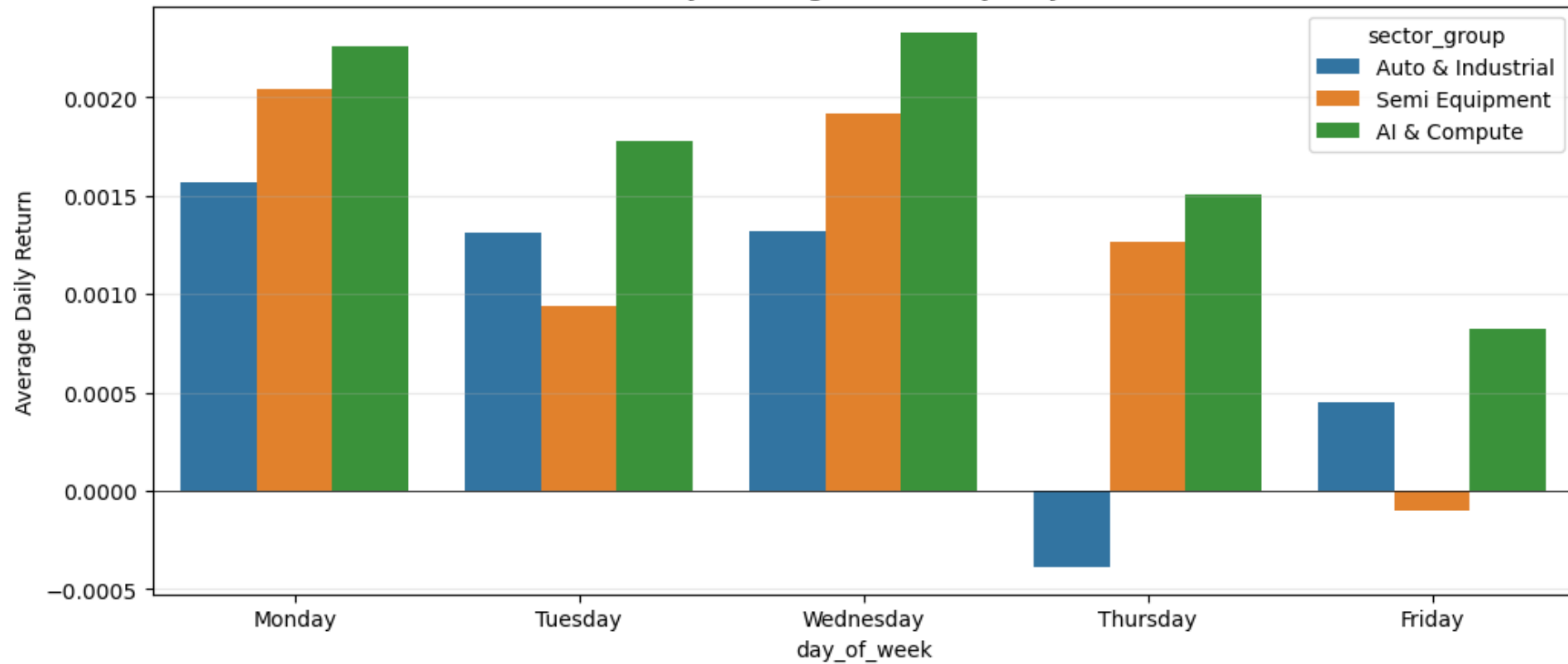
- Dominant supplier of AI training and inference GPUs.
- Direct beneficiary of the post-2022 AI infrastructure build-out.
- Connecting AI software demand & semiconductor supply.
- Used as the primary reference stock.
- Exhibits higher volatility and fatter tails than sector averages.



Seasonality & Trading Behavior

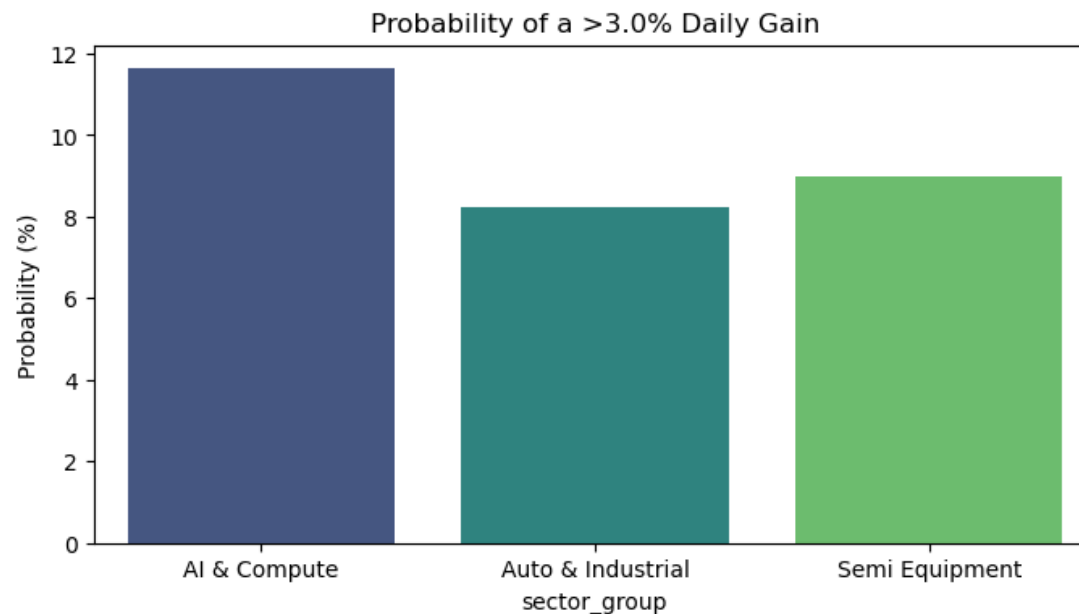
- To check whether return patterns repeat over time.
- Distinguish structural divergence from calendar-driven effects.
- Aggregated daily returns by day of the week.
- Consistent **Friday** dip across all semiconductor sectors.
- Traders de-risk before weekends.
- Short-term behavior remains shared across sectors despite long-term divergence.

Seasonality: Average Return by Day of Week

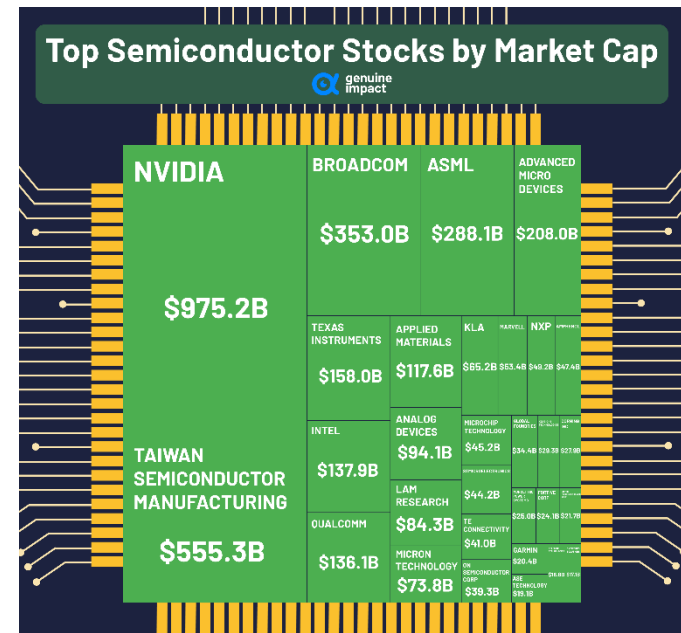


The Concept of Super Days

- A "Super Day" is defined as a day with daily gain $> 3\%$.



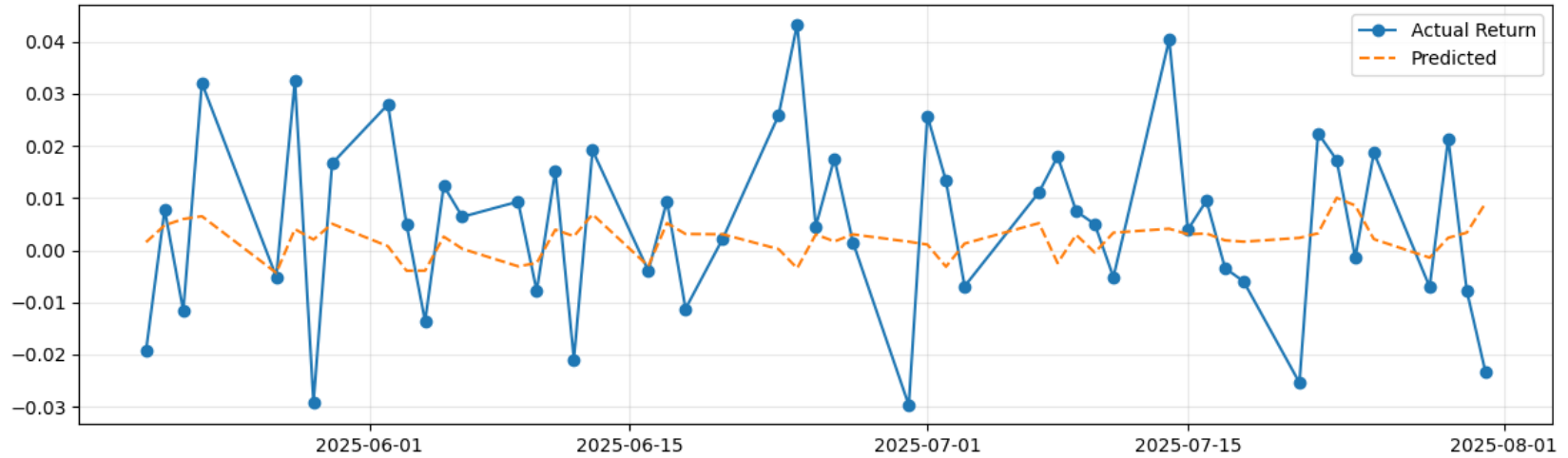
- The probability of a Super Day is **11.6% for AI Stocks** versus only **8.2% for Auto Stocks**.
- AI stocks are growing faster.
- AI stocks are ~40% more likely to produce big jumps.



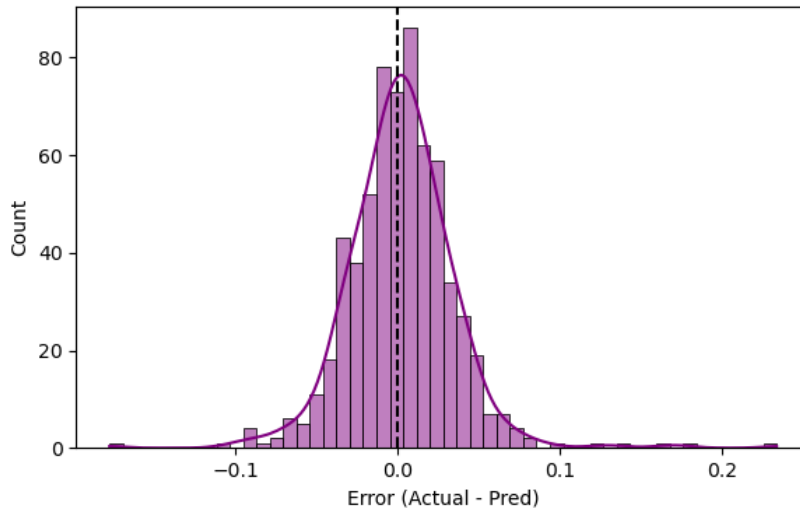
Regression and Predictive Modelling

- Predict NVIDIA (NVDA) return.
- Used time-based train/test split (pre-2023 vs post-2023).
- We compared Linear Regression vs. Polynomial Regression.
- The model's failure is the finding.
- Large under-prediction during 2024 AI rally.
- Historical relationships failed post-AI boom.
- Model failure signals a regime change, not poor modeling.

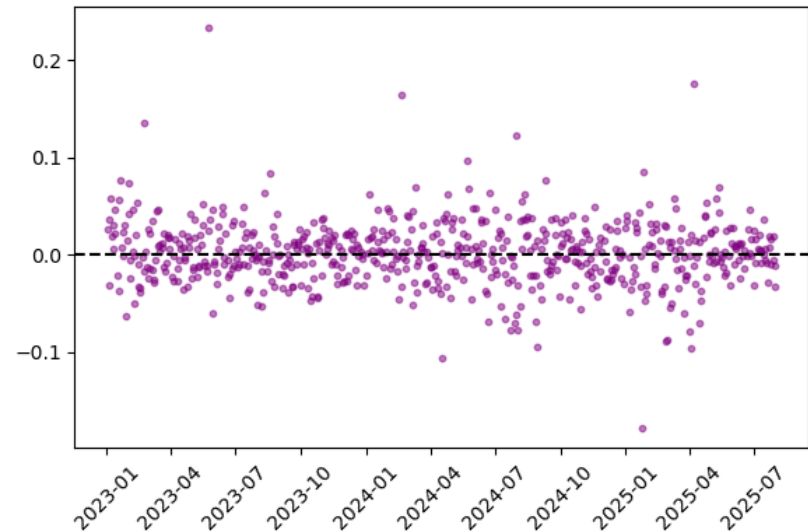
Prediction vs Actual (Last 50 Days)



Residual Distribution (Error)



Residuals Over Time



Key Findings & Conclusions

- The semiconductor industry no longer behaves as a single sector.
- AI & Compute stocks have separated from Auto & Industrial.
- AI stocks exhibit higher growth and higher risk.
- The post-2022 AI boom represents a regime change.
- Models trained on pre-AI data fail to generalize to the new market structure.
- In short, AI did not just accelerate the semiconductor cycle : it rewrote the structure of the industry.